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Skill 33.1 Exercise 1

For each situation, decide whether a Poisson distribution, a binomial distribution, or a normal distribution would be most appropriate. Explain why or why not.

A basketball player takes 12 free throws. We want to model the number of shots they make.

A hospital tracks the number of patients who arrive at the emergency room between midnight and 1:00 AM.

A teacher records the heights of students in a class.

A factory inspects 20 light bulbs and records how many are defective.

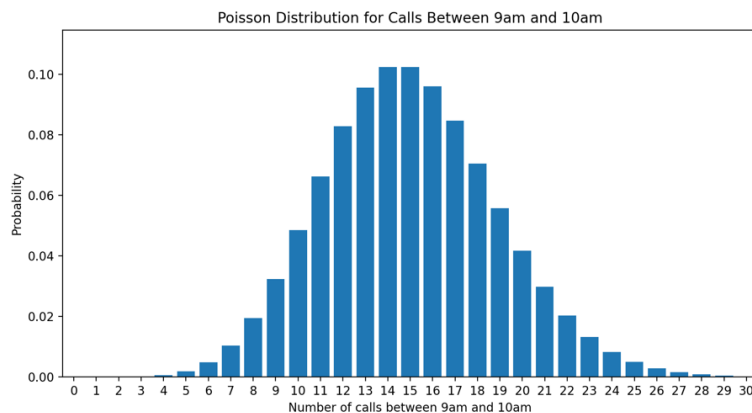
A city records the number of car accidents at a certain intersection each week.

The daily temperatures in a city during the month of April.

Skill 33.2 Exercise 1

The average number of calls in a call center between 9am and 10am is expected to be 15 calls.

Below is the Poisson distribution,



Use the `poisson.pmf()` method below to show how to calculate the probability of getting exactly 15 calls in this time frame.

```
stats.poisson.pmf(k, λ)
```

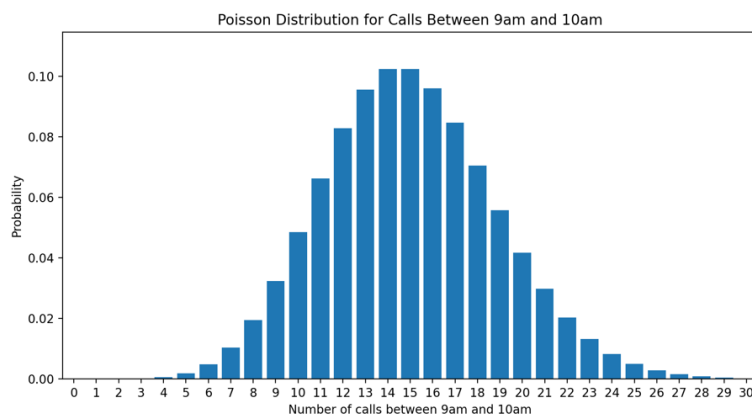
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What is the probability we would get between 7 and 9 calls?

Skill 33.3 Exercise 1

The average number of calls in a call center between 9am and 10am is expected to be 15 calls.

Below is the Poisson distribution,



Use the `poisson.cdf()` method below to show how to calculate the probability of observing more than 20 calls?

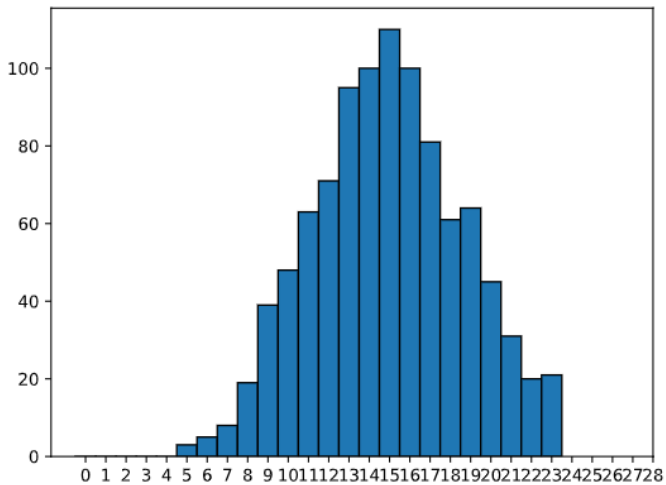
```
stats.poisson.cdf(k, λ)
```

What is the probability of observing between 17 and 21 calls?

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Skill 33.4 Exercise 1

The histogram below shows the distribution of 1000 random variates generated from a Poisson distribution.



Based on the plot, what is the value of lambda, λ ? How do you know?

If we made a plot with a different set of 1000 variates, would it look the same? Explain.

Show how you could use the `poisson.rvs()` method below to simulate 1000 variates with the expected λ value identified above.

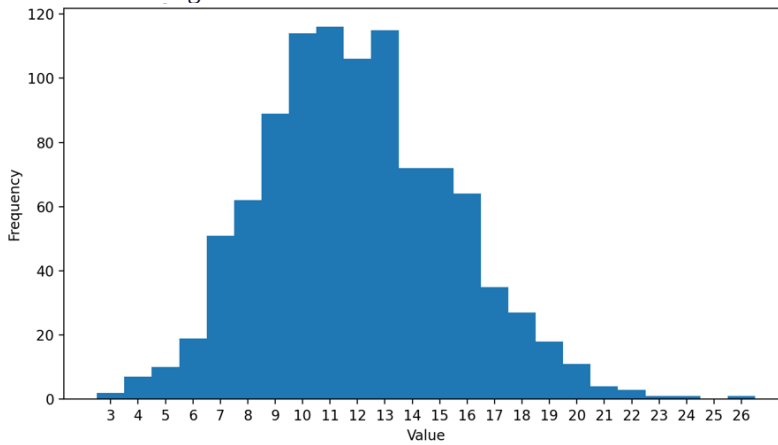
```
rvs = stats.poisson.rvs( $\lambda$ , size = n)
```

What is stored in the variable `rvs`? What is the approximate value of the mean?

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Skill 33.5 Exercise 1

Refer to the histogram below which shows 1000 simulated values from a Poisson distribution.



Based on the plot, what is the value of lambda, λ ? How do you know?

Based on the plot, what is the variance? How do you know?

Show how you could use the `poisson.rvs()` method below to simulate 1000 variates with the expected λ value identified above.

```
rvs = stats.poisson.rvs( $\lambda$ , size = n)
```

Show how you could calculate the variance using the `numpy.var()` method.

```
import numpy as np
```

Skill 33.6 Exercise 1

A certain basketball player has an 85% chance of making a given free throw and takes 20 free throws. What is the expected number of baskets made?

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A student has a 98% chance of arriving early or on time to class. What is the expected number of times the student will arrive LATE in 180 school days?

Skill 33.7 Exercise 1

A certain basketball player has an 85% chance of making a given free throw and takes 20 free throws. Calculate the variance.

A student has a 98% chance of arriving early or on time to class. What is the variance for the number of times the student will arrive LATE in 180 school days?

Skill 33.8 Exercise 1

A machine produces an average of 12 parts in the morning and 9 parts in the afternoon. Let X be the number of parts made in the morning and Y be the number made in the afternoon.

If $E(X)=12$ and $E(Y)=9$, what is $E(X+Y)$?

A student guesses on a 20-question multiple choice quiz with 4 answer choices per question. Let X be the number of correct answers. The expected value is $E(X)=5$.

If the student takes 3 similar quizzes, what is $E(3X)$?

A class's test scores have variance 16. The teacher adds 5 points to every score.

What is the new variance?

Let X be the number of correct answers on a 10-question true/false quiz, where each question is worth 1 point. Pluska decides to weight the test by a factor of 5, what is the new variance?

A student rolls two fair dice. Let X be 1 if the first die lands on six and 0 otherwise, and let Y be 1 if the second die lands on six and 0 otherwise. Find the variance of $X+Y$